Rizwan Flour Mill Rizwan Flour Mill, Islamabad

Report

Project Name	Rizwan Flour Mill
Project Address	Islamabad
Prepared By	Rizwan Salik rizwansalik@hotmail.com

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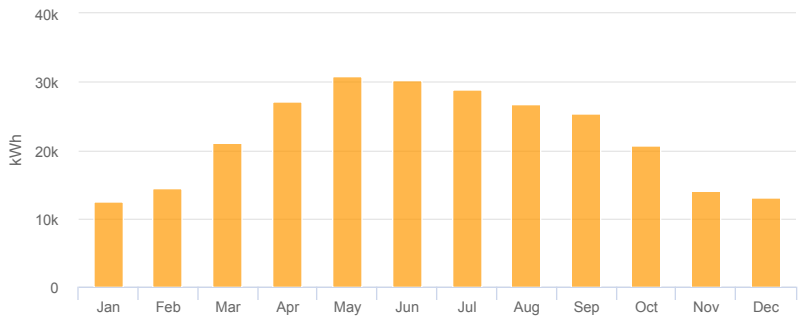
System Metrics

Design	Rizwan Flour Mill
Module DC Nameplate	197.2 kW
Inverter AC Nameplate	230.0 kW Load Ratio: 0.86
Annual Production	265.0 MWh
Performance Ratio	80.9%
kWh/kWp	1,343.7
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)
Simulator Version	9c6cdcd012-87af569316-b27d512130-03c673994f

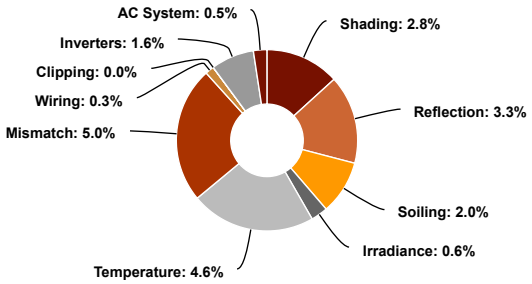
Project Location



Monthly Production



Sources of System Loss



Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,574.9	
	POA Irradiance	1,660.1	5.4%
	Shaded Irradiance	1,614.4	-2.8%
	Irradiance after Reflection	1,561.4	-3.3%
	Irradiance after Soiling	1,530.2	-2.0%
	Total Collector Irradiance	**1,529.8**	**0.0%**
Energy (kWh)	Nameplate	301,697.6	
	Output at Irradiance Levels	299,817.1	-0.6%
	Output at Cell Temperature Derate	285,952.3	-4.6%
	Output After Mismatch	271,525.0	-5.0%
	Optimal DC Output	270,639.0	-0.3%
	Constrained DC Output	270,638.3	0.0%
	Inverter Output	266,308.1	-1.6%
	Energy to Grid	**264,976.5**	**-0.5%**
Temperature Metrics			
	Avg. Operating Ambient Temp		25.2 °C
	Avg. Operating Cell Temp		33.3 °C
Simulation Metrics			
	Operating Hours		4606
	Solved Hours		4606

Condition Set

Description	Condition Set 1											
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)											
Solar Angle Location	Meteo Lat/Lng											
Transposition Model	Perez Model											
Temperature Model	Sandia Model											
Temperature Model Parameters	Rack Type	a	b	Temperature Delta								
	Fixed Tilt	-3.56	-0.075	3°C								
	Flush Mount	-2.81	-0.0455	0°C								
	East-West	-3.56	-0.075	3°C								
	Carport	-3.56	-0.075	3°C								
Soiling (%)	J	F	M	A	M	J	J	A	S	O	N	D
	2	2	2	2	2	2	2	2	2	2	2	2
Irradiation Variance	5%											
Cell Temperature Spread	4° C											
Module Binning Range	-2.5% to 2.5%											
AC System Derate	0.50%											

Module Characterizations	Module	Uploaded By	Characterization
	LR5-72HGD-580M (Longi)	HelioScope	Spec Sheet Characterization, PAN

Component Characterizations	Device	Uploaded By	Characterization
	SUN2000-115KTL-M2 (400V) (Huawei)	HelioScope	Spec Sheet

Components

Component	Name	Count
Inverters	SUN2000-115KTL-M2 (400V) (Huawei)	2 (230.0 kW)
Strings	10 AWG (Copper)	20 (1,816.1 ft)
Module	Longi, LR5-72HGD-580M (580W)	340 (197.2 kW)

Wiring Zones

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	-	5-20	Along Racking

Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Fixed Tilt	Landscape (Horizontal)	Module: 10°	Module: 150°	2.0 ft	4x1	85	340	197.2 kW

Detailed Layout

